

18 *Use of the Data Booklet is relevant to this question.*

25.0 cm³ of hydrochloric acid reacts with 30.0 cm³ of aqueous sodium hydroxide, producing a temperature rise of 28 °C.

What is the heat change that occurs during this reaction?

[Assume density of the solutions = 1.00 g cm⁻³, and that the specific heat capacity of the solutions is the same as the specific heat capacity of water.]

A 2.93 kJ

B 3.51 kJ

C 6.44 kJ

D 69.2 kJ